7.5 Lines and Planes in 3-Space

Introduction In this section we discuss how to find various equations of lines and planes in 3-space.

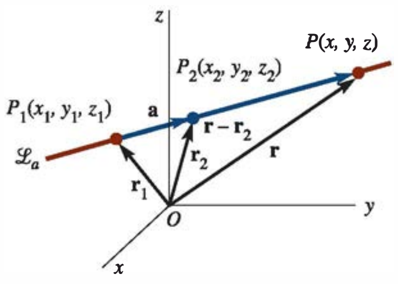

FIGURE 7.5.1 Line through distinct points in 3-space

Lines: Vector Equation As in the plane, any two distinct points in 3-space determine only one line between them. To find an equation of the line through $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$, let us assume that $P(x, y, z)$ is *any* point on the line. In **FIGURE 7.5.1**, if $\mathbf{r} = \overrightarrow{OP}$, $\mathbf{r}_1 = \overrightarrow{OP_1}$, and $\mathbf{r}_2 = \overrightarrow{OP_2}$, we see that vector $\mathbf{a} = \mathbf{r}_2 - \mathbf{r}_1$ is parallel to vector $\mathbf{r} - \mathbf{r}_2$. Thus,

$$\mathbf{r} - \mathbf{r}_2 = t(\mathbf{r}_2 - \mathbf{r}_1). \tag{1}$$

If we write

$$\mathbf{a} = \mathbf{r}_2 - \mathbf{r}_1 = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle = \langle a_1, a_2, a_3 \rangle, \tag{2}$$

then (1) implies that a **vector equation** for the line $\mathscr{L}_a$ is

$$\mathbf{r} = \mathbf{r}_2 + t\mathbf{a}.$$

The scalar t is called a **parameter** and the nonzero vector **a** is called a **direction vector**; the components a_1, a_2, and a_3 of the direction vector **a** are called **direction numbers** for the line.

Alternative forms of the vector equation.

Since $\mathbf{r} - \mathbf{r}_1$ is also parallel to $\mathscr{L}_a$, an alternative vector equation for the line is $\mathbf{r} = \mathbf{r}_1 + t\mathbf{a}$. Indeed, $\mathbf{r} = \mathbf{r}_1 + t(-\mathbf{a})$ and $\mathbf{r} = \mathbf{r}_1 + t(k\mathbf{a})$, k a nonzero scalar, are also equations for $\mathscr{L}_a$.

EXAMPLE 1 Vector Equation of a Line

Find a vector equation for the line through (2, −1, 8) and (5, 6, −3).

SOLUTION Define $\mathbf{a} = \langle 2 - 5, -1 - 6, 8 - (-3) \rangle = \langle -3, -7, 11 \rangle$. The following are three different vector equations for the line:

$$\langle x, y, z \rangle = \langle 2, -1, 8 \rangle + t\langle -3, -7, 11 \rangle \tag{3}$$
$$\langle x, y, z \rangle = \langle 5, 6, -3 \rangle + t\langle -3, -7, 11 \rangle \tag{4}$$
$$\langle x, y, z \rangle = \langle 5, 6, -3 \rangle + t\langle 3, 7, -11 \rangle. \tag{5}$$

≡

Parametric Equations By writing (1) as

$$\begin{aligned}\langle x, y, z \rangle &= \langle x_2 + t(x_2 - x_1), y_2 + t(y_2 - y_1), z_2 + t(z_2 - z_1) \rangle \\ &= \langle x_2 + a_1 t, y_2 + a_2 t, z_2 + a_3 t \rangle\end{aligned}$$

and equating components, we obtain

$$x = x_2 + a_1 t, \quad y = y_2 + a_2 t, \quad z = z_2 + a_3 t. \tag{6}$$

The equations in (6) are called **parametric equations** for the line through P_1 and P_2. As the parameter t increases from $-\infty$ to ∞, we can think of the point $P(x, y, z)$ tracing out the entire line. If the parameter t is restricted to a closed interval $[t_0, t_1]$, then $P(x, y, z)$ traces out a **line segment** starting at the point corresponding to t_0 and ending at the point corresponding to t_1. For example, in Figure 7.5.1, if $-1 \le t \le 0$, then $P(x, y, z)$ traces out the line segment starting at $P_1(x_1, y_1, z_1)$ and ending at $P_2(x_2, y_2, z_2)$.

EXAMPLE 2 Parametric Equations of a Line

Find parametric equations for the line in Example 1.

SOLUTION From (3), it follows that

$$x = 2 - 3t, \quad y = -1 - 7t, \quad z = 8 + 11t. \tag{7}$$

An alternative set of parametric equations can be obtained from (5):

$$x = 5 + 3t, \quad y = 6 + 7t, \quad z = -3 - 11t. \tag{8}$$

≡

Note that the value $t = 0$ in (7) gives $(2, -1, 8)$, whereas in (8), $t = -1$ must be used to obtain the same point.

EXAMPLE 3 Vector Parallel to a Line

Find a vector $\mathbf{a}$ that is parallel to the line $\mathcal{L}_a$ whose parametric equations are $x = 4 + 9t$, $y = -14 + 5t$, $z = 1 - 3t$.

SOLUTION The coefficients (or a nonzero constant multiple of the coefficients) of the parameter in each equation are the components of a vector that is parallel to the line. Thus, $\mathbf{a} = 9\mathbf{i} + 5\mathbf{j} - 3\mathbf{k}$ is parallel to $\mathcal{L}_a$ and hence is a direction vector of the line. ≡

Symmetric Equations From (6), observe that we can clear the parameter by writing

$$t = \frac{x - x_2}{a_1} = \frac{y - y_2}{a_2} = \frac{z - z_2}{a_3}$$

provided that the three direction numbers a_1, a_2, and a_3 are nonzero. The resulting equations

$$\frac{x - x_2}{a_1} = \frac{y - y_2}{a_2} = \frac{z - z_2}{a_3} \tag{9}$$

are said to be **symmetric equations** for the line through P_1 and P_2.

EXAMPLE 4 Symmetric Equations of a Line

Find symmetric equations for the line through $(4, 10, -6)$ and $(7, 9, 2)$.

SOLUTION Define $a_1 = 7 - 4 = 3$, $a_2 = 9 - 10 = -1$, and $a_3 = 2 - (-6) = 8$. It follows from (9) that symmetric equations for the line are

$$\frac{x - 7}{3} = \frac{y - 9}{-1} = \frac{z - 2}{8}.$$ ≡

If one of the direction numbers a_1, a_2, or a_3 is zero in (6), we use the remaining two equations to eliminate the parameter t. For example, if $a_1 = 0$, $a_2 \neq 0$, $a_3 \neq 0$, then (6) yields

$$x = x_2 \quad \text{and} \quad t = \frac{y - y_2}{a_2} = \frac{z - z_2}{a_3}.$$

In this case,
$$x = x_2, \quad \frac{y - y_2}{a_2} = \frac{z - z_2}{a_3}$$

are symmetric equations for the line.

EXAMPLE 5 Symmetric Equations of a Line

Find symmetric equations for the line through $(5, 3, 1)$ and $(2, 1, 1)$.

SOLUTION Define $a_1 = 5 - 2 = 3$, $a_2 = 3 - 1 = 2$, and $a_3 = 1 - 1 = 0$. From the preceding discussion, it follows that symmetric equations for the line are

$$\frac{x - 5}{3} = \frac{y - 3}{2}, \quad z = 1.$$

In other words, the symmetric equations describe a line in the plane $z = 1$. ≡

A line in space is also determined by specifying a point $P_1(x_1, y_1, z_1)$ and a nonzero direction vector $\mathbf{a}$. Through the point P_1, there passes only one line $\mathcal{L}_a$ parallel to the given vector. If $P(x, y, z)$ is a point on the line $\mathcal{L}_a$, shown in **FIGURE 7.5.2**, then, as before,

$$\overrightarrow{OP} - \overrightarrow{OP_1} = t\mathbf{a} \quad \text{or} \quad \mathbf{r} = \mathbf{r}_1 + t\mathbf{a}.$$

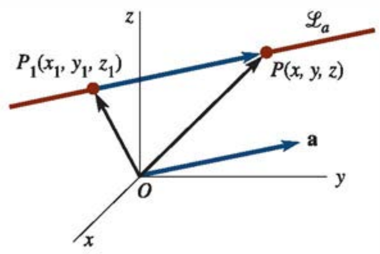

FIGURE 7.5.2 Line determined by a point P and vector $\mathbf{a}$

EXAMPLE 6 Line Parallel to a Vector

Write vector, parametric, and symmetric equations for the line through $(4, 6, -3)$ and parallel to $\mathbf{a} = 5\mathbf{i} - 10\mathbf{j} + 2\mathbf{k}$.

SOLUTION With $a_1 = 5$, $a_2 = -10$, and $a_3 = 2$ we have immediately

$$\textit{Vector:} \quad \langle x, y, z\rangle = \langle 4, 6, -3\rangle + t\langle 5, -10, 2\rangle$$

$$\textit{Parametric:} \quad x = 4 + 5t, \quad y = 6 - 10t, \quad z = -3 + 2t$$

$$\textit{Symmetric:} \quad \frac{x-4}{5} = \frac{y-6}{-10} = \frac{z+3}{2}.$$

≡

Planes: Vector Equation

FIGURE 7.5.3(a) illustrates the fact that through a given point $P_1(x_1, y_1, z_1)$ there pass an infinite number of planes. However, as shown in Figure 7.5.3(b), if a point P_1 and a vector $\mathbf{n}$ are specified, there is only *one* plane $\mathcal{P}$ containing P_1 with **n normal**, or perpendicular, to the plane. Moreover, if $P(x, y, z)$ is any point on $\mathcal{P}$, and $\mathbf{r} = \overrightarrow{OP}$, $\mathbf{r}_1 = \overrightarrow{OP_1}$, then, as shown in Figure 7.5.3(c), $\mathbf{r} - \mathbf{r}_1$ is in the plane. It follows that a **vector equation** of the plane is

$$\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_1) = 0. \tag{10}$$

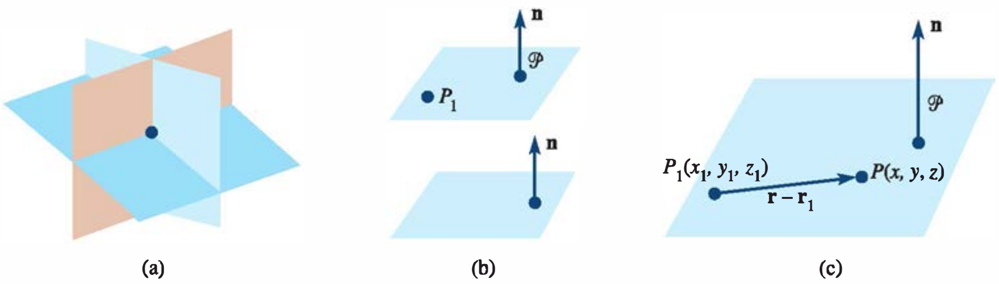

FIGURE 7.5.3 Vector **n** is perpendicular to a plane

Cartesian Equation

Specifically, if the normal vector is $\mathbf{n} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$, then (10) yields a **Cartesian equation** of the plane containing $P_1(x_1, y_1, z_1)$:

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0. \tag{11}$$

Equation (11) is sometimes called the **point-normal form** of the equation of a plane.

EXAMPLE 7 Equation of a Plane

Find an equation of the plane with normal vector $\mathbf{n} = 2\mathbf{i} + 8\mathbf{j} - 5\mathbf{k}$ containing the point $(4, -1, 3)$.

SOLUTION It follows immediately from the point-normal form (11) that an equation of the plane is $2(x - 4) + 8(y + 1) - 5(z - 3) = 0$ or $2x + 8y - 5z + 15 = 0$. ≡

Equation (11) can always be written as $ax + by + cz + d = 0$ by identifying $d = -ax_1 - by_1 - cz_1$. Conversely, we shall now prove that any linear equation

$$ax + by + cz + d = 0, \qquad a, b, c \text{ not all zero} \tag{12}$$

is a plane.

Theorem 7.5.1 Plane with Normal Vector

The graph of any equation $ax + by + cz + d = 0$, a, b, c not all zero, is a plane with the **normal vector** $\mathbf{n} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$.

PROOF: Suppose x_0, y_0, and z_0 are numbers that satisfy the given equation. Then, $ax_0 + by_0 + cz_0 + d = 0$ implies that $d = -ax_0 - by_0 - cz_0$. Replacing this latter value of d in the original equation gives, after simplifying, $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$, or, in terms of vectors,

$$[a\mathbf{i} + b\mathbf{j} + c\mathbf{k}] \cdot [(x - x_0)\mathbf{i} + (y - y_0)\mathbf{j} + (z - z_0)\mathbf{k}] = 0.$$

This last equation implies that $a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$ is normal to the plane containing the point (x_0, y_0, z_0) and the vector $(x - x_0)\mathbf{i} + (y - y_0)\mathbf{j} + (z - z_0)\mathbf{k}$. ≡

EXAMPLE 8 A Vector Normal to a Plane

A vector normal to the plane $3x - 4y + 10z - 8 = 0$ is $\mathbf{n} = 3\mathbf{i} - 4\mathbf{j} + 10\mathbf{k}$. ≡

Of course, a nonzero scalar multiple of a normal vector is still perpendicular to the plane.

Three noncollinear points P_1, P_2, and P_3 also determine a plane.* To obtain an equation of the plane, we need only form two vectors between two pairs of points. As shown in **FIGURE 7.5.4**, their cross product is a vector normal to the plane containing these vectors. If $P(x, y, z)$ represents any point on the plane, and $\mathbf{r} = \overrightarrow{OP}$, $\mathbf{r}_1 = \overrightarrow{OP_1}$, $\mathbf{r}_2 = \overrightarrow{OP_2}$, $\mathbf{r}_3 = \overrightarrow{OP_3}$, then $\mathbf{r} - \mathbf{r}_1$ (or, for that matter, $\mathbf{r} - \mathbf{r}_2$ or $\mathbf{r} - \mathbf{r}_3$) is in the plane. Hence,

$$[(\mathbf{r}_2 - \mathbf{r}_1) \times (\mathbf{r}_3 - \mathbf{r}_1)] \cdot (\mathbf{r} - \mathbf{r}_1) = 0 \qquad (13)$$

is a vector equation of the plane. Do not memorize the last formula. The procedure is the same as (10) with the exception that the vector $\mathbf{n}$ normal to the plane is obtained by means of the cross product.

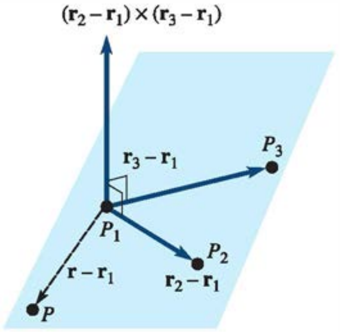

FIGURE 7.5.4 Vectors $\mathbf{r}_2 - \mathbf{r}_1$ and $\mathbf{r}_3 - \mathbf{r}_1$ are in the plane, and their cross product is normal to the plane

EXAMPLE 9 Three Points That Determine a Plane

Find an equation of the plane that contains $(1, 0, -1)$, $(3, 1, 4)$, and $(2, -2, 0)$.

SOLUTION We need three vectors. Pairing the points on the left as shown yields the vectors on the right. The order in which we subtract is irrelevant.

$$\left.\begin{matrix}(1, 0, -1)\\(3, 1, 4)\end{matrix}\right\}\mathbf{u} = 2\mathbf{i} + \mathbf{j} + 5\mathbf{k}, \quad \left.\begin{matrix}(3, 1, 4)\\(2, -2, 0)\end{matrix}\right\}\mathbf{v} = \mathbf{i} + 3\mathbf{j} + 4\mathbf{k}, \quad \left.\begin{matrix}(2, -2, 0)\\(x, y, z)\end{matrix}\right\}\mathbf{w} = (x - 2)\mathbf{i} + (y + 2)\mathbf{j} + z\mathbf{k}.$$

Now, $$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 5 \\ 1 & 3 & 4 \end{vmatrix} = -11\mathbf{i} - 3\mathbf{j} + 5\mathbf{k}$$

is a vector normal to the plane containing the given points. Hence, a vector equation of the plane is $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = 0$. The latter equation yields

$$-11(x - 2) - 3(y + 2) + 5z = 0 \quad \text{or} \quad -11x - 3y + 5z + 16 = 0. \qquad ≡$$

□ **Graphs** The graph of (12) with one or even two variables missing is still a plane. For example, we saw in Section 7.2 that the graphs of

$$x = x_0, \qquad y = y_0, \qquad z = z_0,$$

where x_0, y_0, z_0 are constants, are planes perpendicular to the x-, y-, and z-axes, respectively. In general, to graph a plane, we should try to find

(*i*) the x-, y-, and z-intercepts and, if necessary,
(*ii*) the trace of the plane in each coordinate plane.

A **trace** of a plane in a coordinate plane is the line of intersection of the plane with a coordinate plane.

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*If you ever sit at a four-legged table that rocks, you might consider replacing it with a three-legged table.

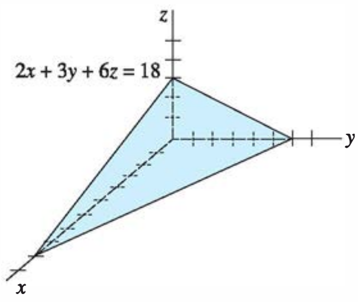

FIGURE 7.5.5 Plane in Example 10

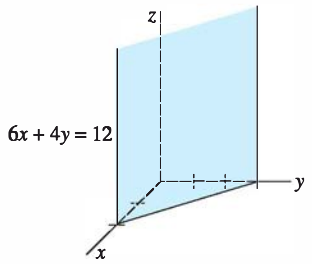

FIGURE 7.5.6 Plane in Example 11

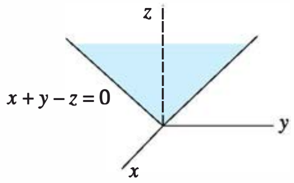

FIGURE 7.5.7 Plane in Example 12

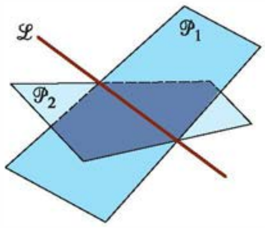

FIGURE 7.5.8 Planes intersect in a line

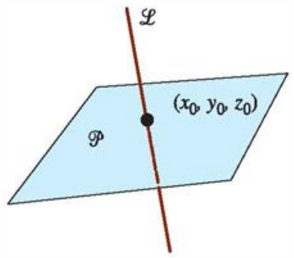

FIGURE 7.5.9 Point of intersection of a plane and a line

EXAMPLE 10 Graph of a Plane

Graph the equation $2x + 3y + 6z = 18$.

SOLUTION Setting:
$$\begin{aligned} y = 0, z = 0 \quad &\text{gives} \quad x = 9 \\ x = 0, z = 0 \quad &\text{gives} \quad y = 6 \\ x = 0, y = 0 \quad &\text{gives} \quad z = 3. \end{aligned}$$

As shown in **FIGURE 7.5.5**, we use the x-, y-, and z-intercepts $(9, 0, 0)$, $(0, 6, 0)$, and $(0, 0, 3)$ to draw the graph of the plane in the first octant. ≡

EXAMPLE 11 Graph of a Plane

Graph the equation $6x + 4y = 12$.

SOLUTION In two dimensions, the graph of the equation is a line with x-intercept $(2, 0)$ and y-intercept $(3, 0)$. However, in three dimensions, this line is the trace of a plane in the xy-coordinate plane. Since z is not specified, it can be any real number. In other words, (x, y, z) is a point on the plane provided that x and y are related by the given equation. As shown in **FIGURE 7.5.6**, the graph is a plane parallel to the z-axis. ≡

EXAMPLE 12 Graph of a Plane

Graph the equation $x + y - z = 0$.

SOLUTION First observe that the plane passes through the origin $(0, 0, 0)$. Now, the trace of the plane in the xz-plane $(y = 0)$ is $z = x$, whereas its trace in the yz-plane $(x = 0)$ is $z = y$. Drawing these two lines leads to the graph given in **FIGURE 7.5.7**. ≡

Two planes $\mathscr{P}_1$ and $\mathscr{P}_2$ that are not parallel must intersect in a line $\mathscr{L}$. See **FIGURE 7.5.8**. Example 13 will illustrate one way of finding parametric equations for the line of intersection. In Example 14 we shall see how to find a point of intersection (x_0, y_0, z_0) of a plane $\mathscr{P}$ and a line $\mathscr{L}$. See **FIGURE 7.5.9**.

EXAMPLE 13 Line of Intersection of Two Planes

Find parametric equations for the line of intersection of

$$\begin{aligned} 2x - 3y + 4z &= 1 \\ x - y - z &= 5. \end{aligned}$$

SOLUTION In a system of two equations and three unknowns, we choose one variable arbitrarily, say $z = t$, and solve for x and y from

$$\begin{aligned} 2x - 3y &= 1 - 4t \\ x - y &= 5 + t. \end{aligned}$$

Proceeding, we find $x = 14 + 7t$, $y = 9 + 6t$, $z = t$. These are parametric equations for the line of intersection of the given planes. ≡

EXAMPLE 14 Point of Intersection of a Line and a Plane

Find the point of intersection of the plane $3x - 2y + z = -5$ and the line $x = 1 + t$, $y = -2 + 2t$, $z = 4t$.

SOLUTION If (x_0, y_0, z_0) denotes the point of intersection, then we must have $3x_0 - 2y_0 + z_0 = -5$ and $x_0 = 1 + t_0$, $y_0 = -2 + 2t_0$, $z_0 = 4t_0$, for some number t_0. Substituting the latter equations into the equation of the plane gives

$$3(1 + t_0) - 2(-2 + 2t_0) + 4t_0 = -5 \quad \text{or} \quad t_0 = -4.$$

From the parametric equations for the line, we then obtain $x_0 = -3$, $y_0 = -10$, and $z_0 = -16$. The point of intersection is $(-3, -10, -16)$. ≡

Remarks

In everyday speech, the words *orthogonal*, *perpendicular*, and *normal* are often used interchangeably in the sense that two objects touch, intersect, or abut at a 90° angle. But in recent years an unwritten convention has arisen to use these terms in specific mathematical contexts. As a general rule, we say that two vectors are *orthogonal*, two lines (or two planes) are *perpendicular*, and that a vector is *normal* to a plane.

7.5 Exercises

Answers to selected odd-numbered problems begin on page ANS-15.

In Problems 1–6, find a vector equation for the line through the given points.

1. $(1, 2, 1), (3, 5, -2)$
2. $(0, 4, 5), (-2, 6, 3)$
3. $(\frac{1}{2}, -\frac{1}{2}, 1), (-\frac{3}{2}, \frac{5}{2}, -\frac{1}{2})$
4. $(10, 2, -10), (5, -3, 5)$
5. $(1, 1, -1), (-4, 1, -1)$
6. $(3, 2, 1), (\frac{5}{2}, 1, -2)$

In Problems 7–12, find parametric equations for the line through the given points.

7. $(2, 3, 5), (6, -1, 8)$
8. $(2, 0, 0), (0, 4, 9)$
9. $(1, 0, 0), (3, -2, -7)$
10. $(0, 0, 5), (-2, 4, 0)$
11. $(4, \frac{1}{2}, \frac{1}{3}), (-6, -\frac{1}{4}, \frac{1}{6})$
12. $(-3, 7, 9), (4, -8, -1)$

In Problems 13–18, find symmetric equations for the line through the given points.

13. $(1, 4, -9), (10, 14, -2)$
14. $(\frac{2}{3}, 0, -\frac{1}{4}), (1, 3, \frac{1}{4})$
15. $(4, 2, 1), (-7, 2, 5)$
16. $(-5, -2, -4), (1, 1, 2)$
17. $(5, 10, -2), (5, 1, -14)$
18. $(\frac{5}{6}, -\frac{1}{4}, \frac{1}{5}), (\frac{1}{3}, \frac{3}{8}, -\frac{1}{10})$

In Problems 19–22, find parametric and symmetric equations for the line through the given point parallel to the given vector.

19. $(4, 6, -7)$, $\mathbf{a} = \langle 3, \frac{1}{2}, -\frac{3}{2} \rangle$
20. $(1, 8, -2)$, $\mathbf{a} = -7\mathbf{i} - 8\mathbf{j}$
21. $(0, 0, 0)$, $\mathbf{a} = 5\mathbf{i} + 9\mathbf{j} + 4\mathbf{k}$
22. $(0, -3, 10)$, $\mathbf{a} = \langle 12, -5, -6 \rangle$
23. Find parametric equations for the line through $(6, 4, -2)$ that is parallel to the line $x/2 = (1 - y)/3 = (z - 5)/6$.
24. Find symmetric equations for the line through $(4, -11, -7)$ that is parallel to the line $x = 2 + 5t, y = -1 + \frac{1}{3}t, z = 9 - 2t$.
25. Find parametric equations for the line through $(2, -2, 15)$ that is parallel to the xz-plane and the xy-plane.
26. Find parametric equations for the line through $(1, 2, 8)$ that is **(a)** parallel to the y-axis, and **(b)** perpendicular to the xy-plane.
27. Show that the lines given by $\mathbf{r} = t\langle 1, 1, 1 \rangle$ and $\mathbf{r} = \langle 6, 6, 6 \rangle + t\langle -3, -3, -3 \rangle$ are the same.
28. Let $\mathcal{L}_a$ and $\mathcal{L}_b$ be lines with direction vectors $\mathbf{a}$ and $\mathbf{b}$, respectively. $\mathcal{L}_a$ and $\mathcal{L}_b$ are orthogonal if $\mathbf{a}$ and $\mathbf{b}$ are orthogonal and parallel if $\mathbf{a}$ and $\mathbf{b}$ are parallel. Determine which of the following lines are orthogonal and which are parallel.
(a) $\mathbf{r} = \langle 1, 0, 2 \rangle + t\langle 9, -12, 6 \rangle$
(b) $x = 1 + 9t,\ y = 12t,\ z = 2 - 6t$
(c) $x = 2t,\ y = -3t,\ z = 4t$
(d) $x = 5 + t,\ y = 4t,\ z = 3 + \frac{5}{2}t$
(e) $x = 1 + t,\ y = \frac{3}{2}t,\ z = 2 - \frac{3}{2}t$
(f) $\dfrac{x + 1}{-3} = \dfrac{y + 6}{4} = \dfrac{z - 3}{-2}$

In Problems 29 and 30, determine the points of intersection of the given line and the three coordinate planes.

29. $x = 4 - 2t,\ y = 1 + 2t,\ z = 9 + 3t$
30. $\dfrac{x - 1}{2} = \dfrac{y + 2}{3} = \dfrac{z - 4}{2}$

In Problems 31–34, determine whether the given lines intersect. If so, find the point of intersection.

31. $x = 4 + t,\ y = 5 + t,\ z = -1 + 2t$
$x = 6 + 2s,\ y = 11 + 4s,\ z = -3 + s$
32. $x = 1 + t,\ y = 2 - t,\ z = 3t$
$x = 2 - s,\ y = 1 + s,\ z = 6s$
33. $x = 2 - t,\ y = 3 + t,\ z = 1 + t$
$x = 4 + s,\ y = 1 + s,\ z = 1 - s$
34. $x = 3 - t,\ y = 2 + t,\ z = 8 + 2t$
$x = 2 + 2s,\ y = -2 + 3s,\ z = -2 + 8s$

The angle between two lines $\mathcal{L}_a$ and $\mathcal{L}_b$ is the angle between their direction vectors $\mathbf{a}$ and $\mathbf{b}$. In Problems 35 and 36, find the angle between the given lines.

35. $x = 4 - t,\ y = 3 + 2t,\ z = -2t$
$x = 5 + 2s,\ y = 1 + 3s,\ z = 5 - 6s$
36. $\dfrac{x - 1}{2} = \dfrac{y + 5}{7} = \dfrac{z - 1}{-1};\quad \dfrac{x + 3}{-2} = y - 9 = \dfrac{z}{4}$

In Problems 37 and 38, the given lines lie in the same plane. Find parametric equations for the line through the indicated point that is perpendicular to this plane.

37. $x = 3 + t,\ y = -2 + t,\ z = 9 + t$
$x = 1 - 2s,\ y = 5 + s,\ z = -2 - 5s;\quad (4, 1, 6)$
38. $\dfrac{x - 1}{3} = \dfrac{y + 1}{2} = \dfrac{z}{4}$
$\dfrac{x + 4}{6} = \dfrac{y - 6}{4} = \dfrac{z - 10}{8};\quad (1, -1, 0)$

In Problems 39–44, find an equation of the plane that contains the given point and is perpendicular to the indicated vector.

39. $(5, 1, 3)$; $2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$
40. $(1, 2, 5)$; $4\mathbf{i} - 2\mathbf{j}$
41. $(6, 10, -7)$; $-5\mathbf{i} + 3\mathbf{k}$
42. $(0, 0, 0)$; $6\mathbf{i} - \mathbf{j} + 3\mathbf{k}$